

TransCIF-FD: Fuel-Decomposition Architecture for Zero-Shot Cross-Region CIF Forecasting

data / physics layer as of FD-16..FD-33 · model frozen (~21k params) · 145-pair LORO: I_{cfg} 40.94 · I_0 37.17 · I_+ 38.2

OUTPUT — 24 h day-ahead CIF trajectory

hourly CIF curve · per-fuel decomposition · ranking signal (Spearman)

3 · INFORMATION TIERS (strictly nested legal inputs)

I_{cfg} **ZERO TELEMETRY (config + weather + calendar)** **40.94 (–35%)**
the deployable tier for telemetry-scarce grids (e.g. Chinese provinces)

I_0 + **336 h renewable-share stream** **37.17 (beats 43.50)**
level anchoring: $\widehat{CIF} + = g((1 - \bar{r}_{48h})ef_{nr} - \overline{\widehat{CIF}})$

I_+ + **observable CIF history** **38.2 (multi-year)**
ZS+ test-time calibration · 6 branches, per-lead inverse-power weights

I_S supervised upper bound (PatchTST, 80% local labels): 43.50

2 · PHYSICS SYNTHESIS (closed form)

$\widehat{CIF}(h) = \sum_f \hat{s}_f(h) ef_f(1 \pm 0.35 \tanh(\cdot))$ **Theorem 1:** $|\widehat{CIF} - CIF| \approx K_{region} |\hat{r} - r| + \varepsilon_{phys}$
bounded EF correction (calibrated EFs, FD-23) · level anchored separately from shape

1 · FuelDecompNet (~21k params — deep conditioning proven harmful ×3)

Solar

astro envelope ×
bounded wx modulation

Wind

IEC wcf / drought-
anchored reference

Baseload ×5

level + support mask
(no phantom fuel)

Thermal ×3

residual + config-
anchored softmax split

Aggregate head

DLinear(rs) + logit
anchor + persist gate

DETERMINISTIC STRUCTURE ROUTING (zero parameters)

$C_{wind} \geq \tau$ or no fuel telemetry → aggregate path · hydro gate $\sigma(20(0.5 - C_{hydro})) \rightarrow$ load-following (BPAT 46.7→16.4)

FEATURE STACK $x_{fuel}(L \times 10) \cdot x_{wx}(L \times 10) \cdot fut_{exog}(H \times 17) \cdot config(16) \cdot ef(10)$

L = 336 h history · H = 24 h horizon · cold-mode dropout p = 0.3 on history → one weight set serves I_{cfg} and I_0

0 · DATA LAYER — public tracks only, no keys

Fuel telemetry — 28 sources

EIA-930 · UK CI API · AU NEMED
DUID (train side only)

Weather — ERA5 + farmblend

capacity-weighted farm weather
(wind R² up to 0.71)

Astro & calendar

solar geometry / clear-sky
UTC-normalised (deterministic)

Day-ahead load forecast

EIA-930, z-scored,
window de-meaned

Fuel prices

WB pink sheets · FRED
(1-month publication lag)

Region config — 16-dim

mean_rs · ef_nr
fuel shares · climate

Curated unit registry

FD-28 — fixes fuel-bucket
labels (hydro in wind)

Calibrated EFs

FD-23 — ridge $\lambda=15$
vs reported CIF, clip [0,1400]

FD-17 wind-speed unit fix (km/h fed to m/s IEC curve; 39–42% of hours misread as cut-out) — largest single gain

source regions (unrestricted) — LORO: 28 sources → 1 unseen target · 29 regions × 5 seeds —

145-PAIR LORO RESULT

median MAE, gCO₂/kWh

I_{cfg} 62.75 → 40.94 (–35%)

I_0 53.39 → 37.17

I_+ 46.9 → 38.2 (multi-year)

beats supervised PatchTST

reference 43.50 by 6.3

win 75% · p = 7.1e–11

TRAINING

$\mathcal{L} = MAE + 1.0 L_{shareEF}$

+ 0.3 L_{rs} + 0.5 L_{shape}

config-distance sampling

$w \propto 1/(|\Delta mean_{rs}| + 0.05)$

600 epochs · Adam 1e–3

all dynamic corrections

zero-initialised (physics warm start)

ALL GAINS = DATA / PHYSICS

model architecture frozen

unit fix · farmblend ×4 rounds

calibrated EFs · registry

deterministic routing

negative probes kept honest:

regime feats, seasonal head,

source bias, shrink λ , demand

ZERO-SHOT DEPLOY (CN)

monthly provincial config →

hourly CIF + fuel decomposition

hydro provinces ~15–19

coal-flat provinces 18–22

(demo: demo_cn_province.py)